

Scope of Work

Project #5: Automation of Steel Beam Calculations (Steel Design)

Client: Kiewit Corporation

Prepared by: Cole Geer, Nathaniel Jones, Tyler Potthoff, Trey Williamson

Course: CIVE 202 – Civil Engineering Analysis II

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Project Background and Purpose:

Project Background

Kiewit Corporation's structural engineering team performs steel member design calculations using the American Institute of Steel Construction (AISC) Manual. While many engineers rely on developed Excel spreadsheets to complete these calculations efficiently, there is a need for standardization across spreadsheets. To address this issue, Kiewit has requested a Python program that automates the design calculations for a W-shaped steel tension member based on AISC Example D.1.

Project Objectives

The primary objective of this project is to translate the steel tension member design calculations from the AISC Manual into a structured and reusable Python workflow that is easy to follow and can be replicated. The program will standardize the calculation process, improve transparency, and demonstrate how engineering design spreadsheets can be automated using code. The secondary objective is to evaluate how changes in input parameters affect design results by analyzing multiple design scenarios.

Scope of Work Tasks

1. Develop Python functions that automate the required calculations for a W-shaped steel tension member in accordance with the AISC Steel Construction Manual.
2. Clearly define all input parameters like material properties, applied loads, etc., and outputs for each function to ensure future users can easily apply the tool.

3. Run three distinct design scenarios by altering input parameters to demonstrate how changes in loading can influence the design results.
4. Create graphical plots to compare the results of the three scenarios and visually show the differences between design cases.
5. Provide written documentation explaining the calculation methodology, function structure, and guidance for future engineers on how to use and modify the Python code.

Deliverables

- A Python notebook that automates W-shaped steel tension member design calculations
- Plots comparing results from three different design scenarios
- A written technical report describing the methodology, inputs, outputs, and application of the automated design tool
- Annotated code document explaining the Python code
- Written report summarizing methods and findings
- GitHub repository containing the notebook and supporting documentation

Data Sources

American Institute of Steel Construction. (2010). *Steel construction manual* (14th ed.). American Institute of Steel Construction.

American Institute of Steel Construction. (2011). *Steel construction manual: Design examples*. American Institute of Steel Construction.